3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs and Expectations
Travelers and Tourists	Discover Egyptian destinations, explore experiences, and create personalized travel plans.
Registered Users	Create memories, share experiences, save journeys, and manage preferences.
Local Residents	Share local knowledge, cultural experiences, and recommendations.
Tourism Businesses	Promote their services and reach potential customers.
System Administrator	Manage users, monitor content, and maintain the platform.
Content Moderators	Review user-generated content and handle inappropriate or misleading submissions.
Development Team	Build, test, document, and maintain the platform.
DEPI Lecturers	Evaluate project documentation, implementation, testing, and presentation.

Note: In the MVP, local residents contribute through the memory feature. Business profiles, offers, and partner dashboards for tourism businesses are planned for a later phase and are therefore not part of the functional requirements below.

3.2 User Stories & Use Cases

User Stories

ID	User Story
US-01	As a user, I want to discover Egyptian destinations so that I can find places to visit.
US-02	As a user, I want to select my mood and interests so that I can discover suitable experiences.
US-03	As a user, I want to view experience details so that I can learn about places and activities.
US-04	As a registered user, I want to create and share memories so that I can document my experiences.
US-05	As a user, I want to explore other users' memories so that I can discover authentic experiences.
US-06	As a user, I want to transform a memory into a personalized travel plan so that I can experience something similar.
US-07	As a user, I want to enter my budget and trip duration so that the AI planner can generate a suitable itinerary.
US-08	As a registered user, I want to save my journey so that I can access it later.
US-09	As an administrator, I want to manage users and content so that the platform remains organized and reliable.

ID	User Story
US-10	As a visitor, I want to register and sign in so that I can create memories and save journeys.
US-11	As a registered user, I want to manage my profile and travel preferences so that recommendations match my interests.
US-12	As a user, I want to see places and experiences on an interactive map so that I can understand where they are located.
US-13	As a registered user, I want to save experiences I like so that I can return to them later.
US-14	As a user, I want to view a traveler's profile and published memories so that I can find authentic recommendations.
US-15	As a content moderator, I want to review user-generated memories so that inappropriate or misleading content is removed.

Use Cases

UC-01: Discover Experiences

Actor: Visitor or registered user.

1. The user opens the discovery page.
2. The system displays available destinations and experiences.
3. The user selects a mood, interest, or category.
4. The system displays matching experiences.
5. The user selects an experience to view its details.

Alternative Flow: If no experiences match the selected mood, interest, or category, the system displays an empty-state message and suggests changing the selection.

UC-02: Create a Memory

Actor: Registered user.

Precondition: The user is signed in.

1. The user opens the memory creation form.
2. The user enters a story and selects a location.
3. The user adds photographs and optional details.
4. The system validates the submitted information.
5. The system saves the memory and displays a confirmation message.

Alternative Flow: If required information is missing or invalid, the system displays validation messages and the memory is not saved.

UC-03: Build an Experience from a Memory

Actor: Registered user.

Precondition: The user is signed in.

1. The user opens a memory.
2. The user selects "Build This Experience."
3. The system requests the budget, duration, travel type, and preferences.
4. The system uses the selected memory and available destination information.
5. The AI planner generates a proposed itinerary.
6. The user reviews and edits the itinerary.
7. The user saves the journey.

Alternative Flow: If the AI planning service is unavailable or fails, the system displays an error message and lets the user try again.

UC-04: Manage Content

Actor: Administrator or content moderator.

Precondition: The user has an administrator or moderator account.

1. The user signs in.
2. The user opens the content management section.
3. The system displays available user-generated content.
4. The user reviews, approves, or removes content.
5. The system saves the changes.

Alternative Flow: If the user is not authorized, the system denies access to the content management section.

UC-05: Register and Sign In

Actor: Visitor.

1. The visitor opens the registration or sign-in page.
2. The visitor enters the required account details or credentials.
3. The system validates the submitted information.
4. The system creates the account or authenticates the user.
5. The system signs the user in and redirects them to the platform.

Alternative Flow: If the information is invalid or the credentials are incorrect, the system displays an error message and the user is not signed in.

3.3 Functional Requirements

Functional requirements describe the features and operations the system must provide.

ID	Functional Requirement	Related Story / Use Case
FR-01	The system shall allow users to register and sign in.	US-10, UC-05
FR-02	The system shall allow users to manage their profiles and travel preferences.	US-11, US-02
FR-03	The system shall display Egyptian destinations and experiences.	US-01, UC-01
FR-04	The system shall allow users to search for and filter experiences by supported categories.	US-01, UC-01
FR-05	The system shall allow users to select moods and interests for experience discovery.	US-02, UC-01
FR-06	The system shall display experience details, including available descriptions, photographs, and locations.	US-03, UC-01
FR-07	The system shall allow registered users to create travel memories, including a story, a place, photographs, and optional details such as visit time, rating, experience type, and tips.	US-04, UC-02
FR-08	The system shall allow users to view published memories.	US-05
FR-09	The system shall allow users to associate memories with specific places.	US-04, UC-02
FR-10	The system shall allow users to select a memory as the starting point for a personalized experience.	US-06, UC-03
FR-11	The system shall collect travel-planning information, including budget, duration, destination, and travel preferences.	US-07, UC-03
FR-12	The system shall generate a proposed itinerary using the AI planning service, with a short explanation of why each activity was suggested.	US-06, US-07, UC-03
FR-13	The system shall allow registered users to save and view their journeys.	US-08, UC-03
FR-14	The system shall allow users to organize itinerary activities by day, time, and location.	US-08, UC-03
FR-15	The system shall allow authorized administrators and moderators to manage users and moderate content.	US-09, US-15, UC-04
FR-16	The system shall validate user inputs and display appropriate success and error messages.	All stories
FR-17	The system shall display places and experiences on an interactive map.	US-12
FR-18	The system shall allow registered users to save experiences to a personal list.	US-13
FR-19	The system shall allow users to view a traveler's public profile and published memories.	US-14

3.4 Non-Functional Requirements

Non-functional requirements define the quality and operational standards of ECHO Egypt.

Performance

- The system should load main pages efficiently under normal operating conditions.
- Common API requests should target a response time of two seconds or less under agreed test conditions, excluding external AI processing and large media transfers.
- Images and database queries should be optimized to reduce loading time.

Security

- The system shall protect user accounts through secure authentication.
- The system shall enforce authorization for protected operations.
- The system shall validate user inputs on the server.
- Sensitive credentials and API keys shall not be exposed in public code.
- Users shall not be allowed to modify other users' private data without authorization.
- Uploaded images shall be validated for file type and size.

Usability

- The interface shall be responsive across desktop, tablet, and mobile devices.
- Navigation and user actions shall be clear and consistent.
- Forms shall provide understandable validation messages.
- The system shall display appropriate loading, success, empty, and error states.

Reliability

- The system shall handle database and external API failures gracefully.
- Saved memories and journeys shall remain available after application restarts when successfully stored in persistent storage.
- The system shall provide useful error messages when an operation fails.
- The system should support data backup and recovery procedures.
- The deployed system should remain accessible during the evaluation and demonstration period.

Maintainability

- The source code shall follow a consistent structure and naming convention.
- The system should use reusable components and modular code.
- Configuration and deployment instructions shall be documented.

Compatibility

- The system shall support current stable versions of the selected major web browsers.
- The interface shall function correctly on supported desktop and mobile screen sizes.